BIOMIMETIC ML DIAGNOSTICS

Heart Disease Predictor Application

Precision cardiovascular risk classification powered by machine learning and hosted on Streamlit Cloud.



Streamlit Engine • 13 Clinical Biomarkers

PROJECT CONTRIBUTORS

Core Development Team & Mentorship

PROJECT GUIDE
Shekhar Kumar

TEAM LEAD ID: 6605603

Vishal Kumar Shrivastava

Full-Stack ML Engineer

- End-to-end system design & integration
- Model training, validation & serialization
- Streamlit UI design & cloud deployment

MEMBER ID: 6605605

Abhinav Saraf

Documentation Lead

- Project report & presentation compilation
- Architecture workflow documentation
- Technical specifications lead

MEMBER ID: 6606824

Anurag Yadav

ML Model Trainer

- Dataset preprocessing & feature scaling
- Model evaluation & accuracy benchmarking
- Hyperparameter tuning & optimization

SYSTEM HIGHLIGHTS

Application Core Capabilities

Real-Time Scoring

Instant cardiovascular risk prediction calculated under 50ms per patient entry.

Multi-Model Comparative

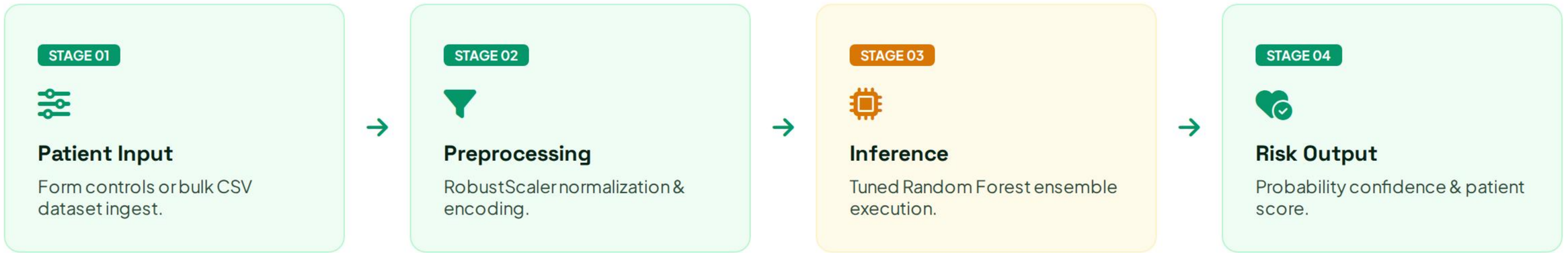
Cross-verifies individual results across 4 pre-trained classification algorithms.

Batch Assessment

Seamless CSV batch upload for cohort-wide screening and bulk export.

DATA FLOW FLOWCHART

End-to-End Prediction Architecture



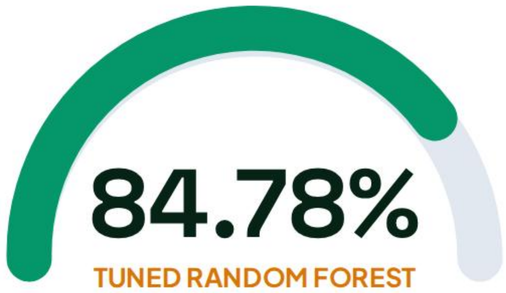
FEATURE ENGINEERING

Key Biomarkers & Clinical Features

Biomarker Label	Clinical Variable Name	Reference Scale / Units
Chest Pain Severity	cp	0: Asymptomatic to 3: Typical Angina
Resting Blood Pressure	trestbps	90 - 200 mm Hg (On Admission)
Serum Cholesterol	chol	126 - 564 mg/dl
Max Exercise Heart Rate	thalach	71 - 202 bpm
ST Depression (Exercise)	oldpeak	0.0 - 6.2 Relative to Rest

MODEL BENCHMARK

Peak Diagnostic Accuracy



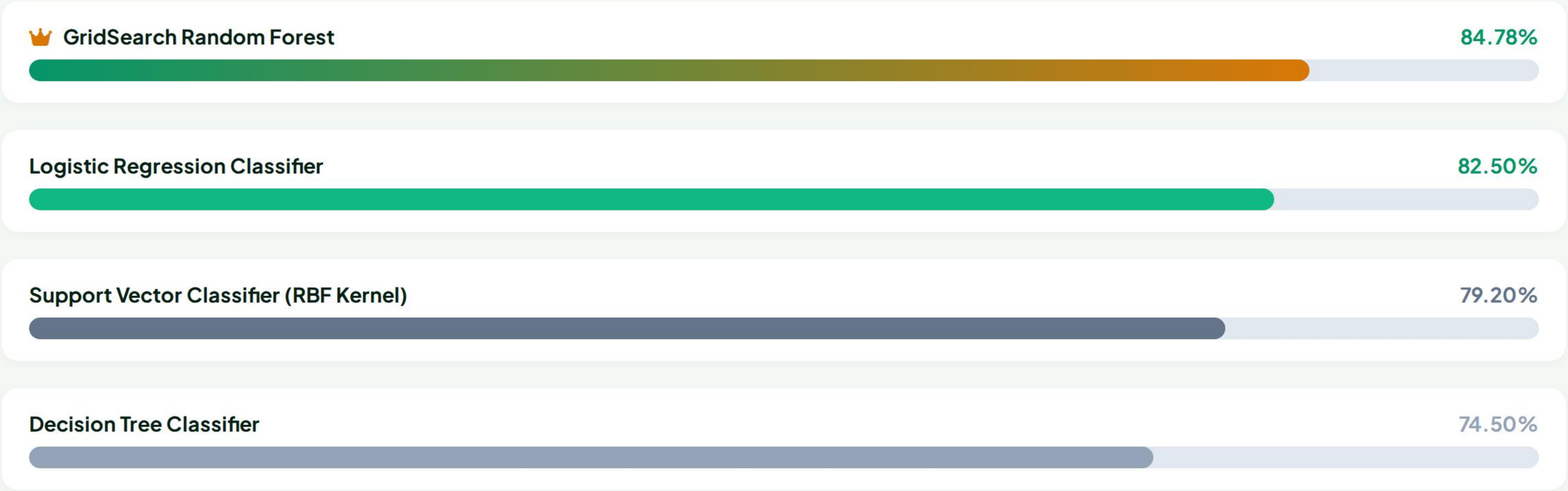
Cross-Validated Consistency

Achieved through 10-fold cross-validation grid search optimizing estimator trees and max depth.

Precision	Recall
86.2%	83.5%

ALGORITHM EVALUATION

Model Performance Leaderboard



USER EXPERIENCE

Streamlit App Interface Highlights

Dynamic Controls

Sliders & selection boxes matching exact diagnostic bounds.

Probability Metric Cards

Displays continuous probability score percentage with color severity.

STREMLIT DEPLOYMENT SPECS

- ✓ Low latency browser execution
- ✓ Zero database dependence for inference
- ✓ Responsive layout for mobile & tablet

GOVERNANCE

Clinical Deployment Safeguards

Augmentative Tool

Assists healthcare professionals; does not replace qualified diagnostic judgment.

Stateless Evaluation

Patient clinical inputs are never retained or saved on public cloud memory.

ROADMAP

Future Enhancements

XAI Integration

Implement SHAP waterfall plots for patient-specific feature explanation.

EHR Integration

HL7 / FHIR API connector support for hospital systems ingest.



Heart Disease Predictor

Empowering clinical screening through machine learning precision.

</> Python & Scikit-Learn Streamlit Cloud